

21

Acquisition of Suprasegmental Phonology in Adult Bilingualism

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21.1 Introduction

This chapter reviews the last twenty-five years of L2 (second language) prosody research in three sections – word stress (Section 21.2), sentence intonation (Section 21.3), and rhythm (Section 21.4) – and presents findings in relation to two underlying themes: form-meaning mapping and bilingual contexts. While my selection of section topics is somewhat arbitrary and nonexhaustive by necessity, given the large scope of L2 prosody, I have chosen those two underlying themes because they illustrate how L2 prosody research has addressed issues of current concern. Second language research is conditioned by the fact that English has become the *lingua franca par excellence*, a status that represents something of a two-edged sword (Soler & Morales-Galvez, 2022). While on the one hand it facilitates transnational communication, on the other it is linked to language subordination ideologies (Lippi-Green, 2012). Consequently, a researcher's approach to L2 prosody will inevitably favor a particular ideology. For example, focusing L2 English research only on educated US or British varieties ignores the fact that English is native to millions of speakers in South Asia and thus indirectly reinforces a colonial frame of mind (Kachru, 1992; Soler & Morales-Galvez, 2022). As a result, L2 researchers (e.g., Anya, 2021; Levis, 2020; Rosa & Flores, 2017) are increasingly adopting approaches that incorporate social justice and equity into their work, including research in L2 prosody (Armstrong et al., 2022).

As we will see, social factors interact with linguistic and cognitive factors in the assigning of meaning to sentence intonation patterns. A brief example illustrates this point. Sociolinguist John J. Gumperz (1997) noted that newly hired South Asian women working at a cafeteria in a major British airport were perceived as rude by British customers because when they offered “Gravy?” they used a falling intonation instead of the rising one British service staff would have used. This cross-cultural misinterpretation of sentence intonation that is perfectly polite in Indian English emerged in a specific language contact context with clear power

differentials between interlocutors in terms of gender, profession, and national origin. If the context were instead a university cafeteria in India and a group of local students invited a British exchange student to take a seat at their table using precisely the same falling intonation, the young visitor would be unlikely to take offense. Although mapping meaning to an intonation form is unarguably linked to linguistic and cognitive factors, it cannot be analyzed separately from the context of language contact in which it takes place as well as the analyst's own sensitivity to social patterns and individual attitudes.

I therefore endeavor in this chapter to apply each topic covered to contexts of both *additive bilingualism*, where the L2 is added to the L1 and thus the learner's total linguistic capital, and *subtractive bilingualism*, where L2 acquisition takes place at the expense of the L1 and thus makes no net contribution to the learner's total linguistic capital. In additive bilingualism, the L2 is acquired in an educational setting and within a society that acknowledges and values the use of the learner's L1 and regards acquisition of that particular L2 as enhancing the learner's status. In other words, this bilingualism occurs in a context of privilege. This sort of bilingualism is exemplified by the acquisition of English by the educated middle class in much of Latin America, where children learn their first language, Spanish, at home and become literate in that language at school before starting to learn their L2, English. Subtractive bilingualism, on the other hand, occurs when the learner acquires an L2, which is attributed higher prestige by the broader community, at the expense of their L1. Native speakers of the indigenous languages of Latin America, for example, know that mastery of the registers of the Spanish they learn in school is more likely to confer social advantage than an enriched command of their L1, regardless of the prestige ascribed to the L1 within their home community (Flores & García, 2020).

Our discussion of L2 prosody will also require the use of specialized terminology. First of all, an utterance can be broken down into a hierarchy of prosodic units, the above-word level, the word level, and the below-word level, as illustrated in Figure 21.1 for the Catalan proverb *Com més*

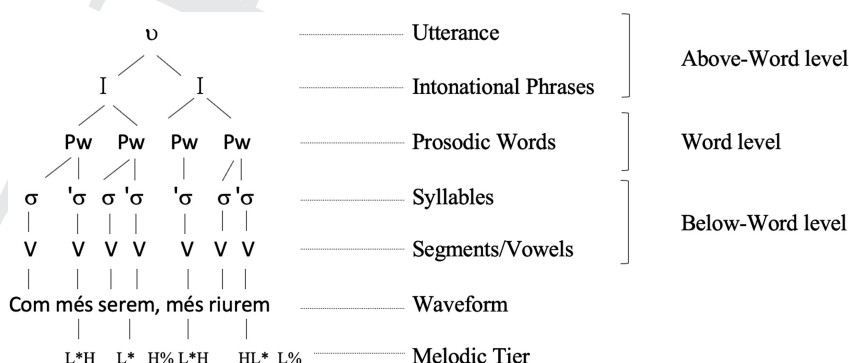


Figure 21.1 The prosodic hierarchy. Prosodic domains are hierarchically organized. As a stress-accent language, the melodic tier in Catalan attaches to stressed vowels and the right edge of intonational phrases.

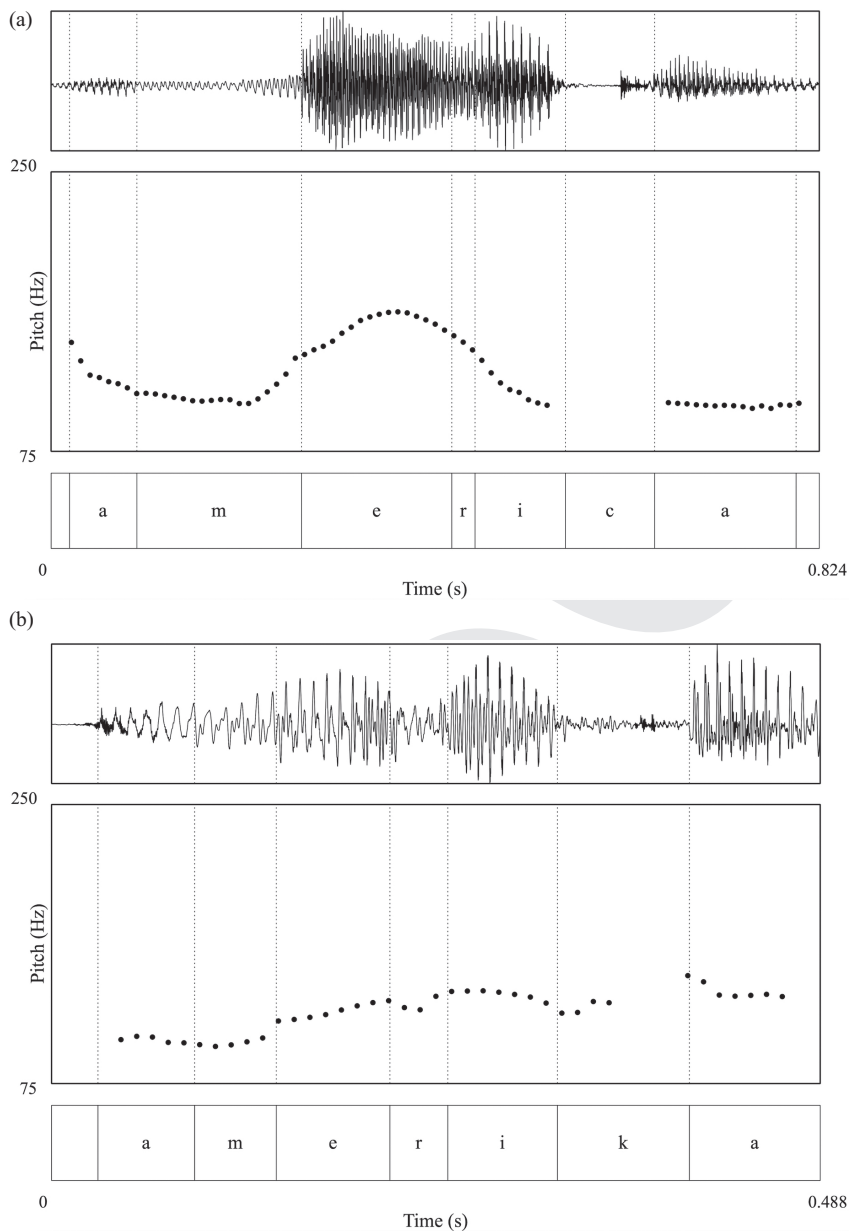


Figure 21.2 Acoustic cues to English stress (a) and Japanese pitch accent (b) in the cognate "America." Pitch and duration cue stress in English, while only pitch – the ascending F0 on "e" – cues the Japanese pitch accent.

serem, més riurem ("The more [people] we are, the more we will laugh" [i.e., the more, the merrier]). At the word level, languages give perceptual prominence to a syllable or mora by enlarging its duration and intensity, like stress in English and Spanish, or by varying its pitch (i.e., frequency), like a pitch accent in Tokyo Japanese (Figure 21.2). Stress, pitch accents, and tone allow

languages to distinguish between two otherwise identical forms, known as word-minimal pairs. For example, stress on the first syllable distinguishes the English noun “object” from the second-syllable-stressed verb “object.” Similarly, Japanese pitch accented *hashi* (“chopsticks”) contrasts with unaccented *hashi* (“edge”), while in Chinese *ma* uttered with tone 1 means “mother” but with tone 2 it means “numb.” Languages with fixed stress, however, lose this contrastive function: the fixed word-initial stress of Hungarian, for example, demarcates word beginnings but cannot generate word-minimal pairs.

Domains at the above-word level relate to sentence intonation. The right and/or left edges of these domains are marked with boundary tones, such as the H% and L% in the “melodic tier” in Figure 21.1. They also contain pitch movements related to lexical events such as stress, pitch accent, and tone. Languages with above-word domains containing both boundary tones and pitch movements related to lexical prominence are called *head-languages*. By contrast, in what are known as *edge-languages* these domains have only boundary tones. Within head-languages, the pitch contour of above-word domains may be either fixed, as in Tokyo Japanese, where a hat shape repeats on each accentual phrase, or variable, as in stress-accent languages like English, where the stressed syllable’s pitch shape is contingent on the particular meaning that sentence intonation is intended to convey.

Below-word domains are related to rhythm. Rhythm is the sensation of isochrony (the division of time into equal portions) conveyed by the repetition of speech units like syllables, morae, or lexical stress. This perceptual isochrony has been thoroughly studied in relation to duration correlates, but only recently have intensity and pitch been investigated and related to word and larger domains.

The remainder of the chapter is organized as follows. Section 21.2 on L2 stress presents pioneering research (Section 21.2.1) which was later extended to other forms of word prominence such as tone and pitch accents, and to other edge-languages besides French mainly in additive bilingualism contexts (Section 21.2.2). Relevant findings in contexts of subtractive bilingualism are summarized in Section 21.2.3. Section 21.3 reviews L2 sentence intonation. It introduces the tools most widely used to analyze sentence intonation and the many sources of variation they capture (Section 21.3.1). Two sources of this variation are discussed, namely typological diversity in expressing focus meaning (Section 21.3.2) and accommodation patterns in situations of subtractive bilingualism (Section 21.3.3). Section 21.4 reviews rhythm, first defining the concept (Section 21.4.1) and then summarizing the methods by which it can be measured (Section 21.4.2). Research findings from contexts of additive (Section 21.4.3) and subtractive language contact (Section 21.4.4) are then presented. Conclusions and directions for future research (Section 21.5) end the chapter.

21.2 L2 Stress

21.2.1 Early Research

Early research into L2 stress was guided by two unexpected findings. First, it was found that native English speakers serving as controls were outperformed by Dutch speakers of L2 English in a stress-matching task (Cooper, Cutler, & Wales, 2002). After listening to the first syllable of English words such as “music” and “museum” – where lexical stress is unusually expressed by only suprasegmental cues (i.e., duration, pitch, and intensity) and not by vowel reduction – participants were asked to match stressed [mju] to [‘mjusik] and unstressed [mju] to [mju’sium]. The Dutch-speaking participants were faster and more accurate at doing this than the native speakers. This unexpected result was framed as a case of L1-to-L2 transfer of suprasegmental cues (Cutler, 2005, 2012; Cutler and Jesse, 2021 and references therein). In general, vowel reduction is a more reliable segmental cue to lexical stress in English than it is in Dutch. Consequently, English speakers rely mostly on vowel reduction to perceive stress in English whereas Dutch speakers rely on both vowel reduction and suprasegmental cues to process Dutch stress. When listening to English stress, the Dutch speakers transferred their L1 reliance on suprasegmental cues, resulting in a superior performance in that specific context. More recently, Bruggeman and colleagues have explored whether the amount of L2 input modulated this initial L1-to-L2 cue transfer (Bruggeman, 2016; Bruggeman, Yu, & Cutler, 2022). Using the same experimental design and materials, they tested L1 Dutch speakers who had been living in Anglophone Canada for at least thirty years. For these speakers, vowel reduction cues outweighed suprasegmental cues, matching native English speakers’ performance. Thus, while L1 cue transfer into L2 explained L2 stress at the initial stages of L2 acquisition, at later stages and given enough L2 input, this initial transfer was readjusted to optimize efficiency in L2 processing.

In the second study yielding unexpected results, advanced-level French speakers of Spanish – despite their high proficiency – experienced persistent “stress deafness” in tasks requiring abstraction such as sequence recall and lexical decision tasks. However, they did perceive stress in tasks involving phonetic perception with minimal abstraction, such as AX (same/different) discrimination tasks (Dupoux et al., 1997; Dupoux, Peperkamp, & Sebastián-Gallés 2001, 2010; but see Llisterri and Schwab [2019] and references therein for differing interpretations). Similar results were obtained from speakers of languages with fixed stress such as Polish and Hungarian (Peperkamp, Vendelin, & Dupoux, 2010). These speakers’ stress deafness was interpreted as a processing deficit linked to their inability to represent contrastive stress in their L2 Spanish lexicon. Because these speakers matched form – that is, suprasegmental cues to a demarcative meaning such as boundary tones in accentual phrases in French and fixed stress in Hungarian – learning to perceive L2 Spanish stress implied remapping suprasegmental

cues from their L1 demarcative meaning to the contrastive meanings conveyed by stress in Spanish.

Summarizing, the persistent stress deafness experienced by French speakers of Spanish was related to the lack of stress representation in their L2 lexicon, while Dutch speakers' outstanding perception of L2 stress was framed as an initial L1 cue transfer with later cue reweighting to match the target language. These different approaches came from the comparison on the one hand of two prosodically similar languages, Dutch and English – both head-languages with contrastive stress – and on the other of two prosodically different languages, French, which has no word prominence, and Spanish, a head-language with contrastive stress. Recent research continues to disentangle the effect of L1–L2 prosodic distance outcomes.

21.2.2 Recent Research

21.2.2.1 Adding Tone and Pitch Accents to L2 Stress

Recent research has contributed to the discussion in Section 21.2.1 by augmenting the number of L1–L2 language pairs and methodologies examined and has revolved around two questions. The first concerns the degree to which speakers of a head-language learning a second head-language with a different type of word prominence – be it contrastive stress, tone, or pitch accent – transfer the prominence type of their L1 to their L2. For example, Wiener and Goss (2018, 2019) examined pitch accent perception in L2 Tokyo Japanese by native English (stress) and Mandarin (tone and stress) speakers. Mandarin speakers with no prior knowledge of Japanese outperformed English speakers with an intermediate level of Japanese, and English speakers who had previously studied Mandarin outperformed English speakers who had not, showing L1-to-L2 transfer between L1 stress and tone to L2 pitch accents. Moreover, prior experience with more than one word-prominence type (e.g., stress and tone) facilitated the perception of L2 pitch accents to a larger extent than prior experience with a single prominence type (e.g., stress only).

In relation to acoustic cues to word prominence, Qin, Chien, and Tremblay (2017) showed that only cues that were already used in the learners' L1 were transferred to the L2. They compared the perception of English stress by Standard Mandarin (tone and stress) with Taiwan Mandarin speakers (tone and no stress) in natural and synthesized speech. Their results showed that when they listened to natural speech the fact that tone was used in their L1s helped speakers from both groups to perceive stress in English. However, when exposed to the synthesized stimuli, Standard speakers used both pitch and durational cues while Taiwan speakers relied only on pitch, indicating that speakers' perception of English stress was based on the cues that signaled word prominence in their own L1s. A more recent study by Choi (2022) showed that functional load also modulated L1-to-L2 cue transfer. Thus, while Cantonese learners of English attended in general to pitch cues to English stress,

confirming L1 transfer, they switched to vowel-quality cues in contexts where vowel quality was more informative than pitch.

The second question of interest in research on the acquisition of L2 stress concerns whether a native representation of word prominence is required to learn word prominence in an L2. This question has been explored by comparing the performance of native speakers of head- and edge-languages as they learned an L2 that was a head-language. Most studies have examined the perception of L2 English stress by speakers of different L1s, yielding consistent results (e.g., Choi, Tong, & Samuel, 2019; Kim & Tremblay, 2021; Lin et al., 2014). Native speakers of head-languages outperform those of edge-languages regardless of the type of word prominence characterizing their L1. For example, native Mandarin speakers (a tone language with stress) outperformed native Seoul Korean speakers (an edge-language) in perceiving English stress in sequence recall and lexical decision tasks (Lin et al., 2014). Similarly, speakers of Gyeongsang Korean (a pitch accent language) outperformed Seoul Korean speakers (Kim & Tremblay, 2021).

21.2.2.2 Adding Pitch Shape and Post-Lexical Processing to L2 Stress

As noted in Section 21.2.1, early research related the persistent stress deafness experienced by French speakers of Spanish to the lack of stress representation in their L1 lexicon. By contrast, native speakers of head-languages with unpredictable stress, such as English or Spanish, which represent stress in their L1 lexicons, were assumed to perceive L2 stress accurately across the board. Recent research has qualified this assumption by showing gradient differences in stress perception depending on the L1 of the listener. Thus, Ortín and Simonet (2022, 2023) showed that, at the pre-lexical level, Spanish speakers were more accurate than English speakers at perceiving stress in nonwords (e.g., “númi” versus “numi”) in sequence recall tasks. By the same token, at the post-lexical level, Ortega-Llebaria, Gu, and Fan (2013) found that Spanish speakers more accurately perceived stress in tasks involving Spanish target words with controlled cues embedded in sentences with different intonations from English speakers. The authors concluded that the English speakers’ reliance on pitch cues explained their perception of paroxytone words as oxytones in declarative sentences (e.g., [‘mama] perceived as [ma’mɑ]), while they failed to perceive stress altogether in flat-F0 parenthetical clauses. Taken as a whole, these results show that, within head-languages, L2 stress perception can be gradient at different levels (i.e., pre-lexical versus post-lexical) of processing.

Many studies have examined stress perception in words with an F0-peak on the stressed syllable (i.e., H*). Nevertheless, the pitch shape of stress is variable and contingent on sentence intonation requirements, so that, for example, the stressed syllable “ma” bears an F0-peak in “Mary!” but an F0-valley in “Mary?,” while the pitch shapes of tones and pitch accents are fixed in the lexicon, as in Mandarin *ma1* “mother” versus *ma2* “numb.” Consequently, native speakers of tonal languages tend to overlook variation in the pitch shape of stress in an L2.

Recent research has revealed such effects in English word recognition by bilingual L1 Mandarin-L2 English speakers (Ortega-Llebaria, Nemogá, & Presson, 2017; Shook & Marian, 2016; Wang, Wang, & Malins, 2017). While both monolingual English controls and Mandarin-English bilinguals processed pitch shape at the pre-lexical level to normalize, for example, for gender differences, only Mandarin-English bilinguals continued processing pitch shape at the lexical level. Crucially, these speakers organized both the Mandarin and the English words in their bilingual lexicon with regard to pitch shape as if English was a one-tone language (Ortega-Llebaria & Wu, 2021; Wang, 2021). Nonetheless, future L2 research needs to further compare the effects of pitch shape on the perception of L2 word prominence in speakers of stress languages and speakers of tone languages.

21.2.3 L2 Stress in Subtractive Bilingualism: The Case of Heritage Spanish Speakers

Since the patterns described so far have been mostly documented in contexts of additive bilingualism, there is always the risk that any conclusions to be drawn do not reflect what occurs in the context of subtractive bilingualism. The body of research on the perception of stress by heritage speakers of Spanish sheds considerable light on this issue. The well-documented difficulties that heritage speakers encounter with Spanish stress are not orthographic but rather related to locating stress in the correct syllable of a word (e.g., Parera & Lynch, 2021). Importantly, Kim (2020) showed an incongruent pattern between the production and the perception of Spanish stress by heritage speakers that was not present in either native Spanish speakers or English speakers of L2 Spanish. That is, while native Spanish speakers displayed ceiling accuracy in both perception and production tasks, English speakers of Spanish were equally inaccurate in both skill areas. By contrast, heritage speakers displayed high accuracy in perception but low accuracy in production. This incongruence was related to the context in which subtractive bilingualism occurs: heritage speakers inhabit an environment that favors the development of a perceptual representation of stress (i.e., continual receptive exposure to Spanish at home) but insufficient contexts of use to develop stable production patterns (i.e., at school English is the vehicular language). Moreover, Ortín (2022) showed that heritage speakers' perception of Spanish stress was heavily modulated by language dominance and their knowledge of Spanish. The more heritage speakers used Spanish and the more proficient they became, the closer their performance was to that of native Spanish speakers. These results provided supporting evidence for earlier research (Rao, 2019; Saalfeld, 2012) arguing that the knowledge of Spanish acquired by heritage speakers through an early exposure to Spanish at home can be lost if this use is interrupted early in development.

21.2.4 Summary

Difficulties of L2 stress perception were initially framed either as an L1-to-L2 cue transfer problem or as a processing deficit linked to learners' inability to represent contrastive stress in their lexicons. Recent research has established the extent and limits of those initial frames by examining a larger number of L1–L2 language pairs. Native speakers of head-languages tend to experience fewer difficulties than native speakers of edge-languages in perceiving L2 stress, pitch accents, and tone, suggesting that L1 representations of word prominence help them to learn any word-prominence type in an L2. However, the gradient perception displayed by native speakers of different head-languages suggests that an L1 word-prominence representation is a necessary – albeit insufficient – condition for the accurate perception of word prominence in an L2. In head-languages, only the cues to word prominence used in the L1 are transferred to an L2, thus modulating L2 perception. While these patterns are reported to occur in contexts of additive bilingualism, in subtractive bilingualism contexts an asymmetry has been found between the perception and the production of stress that is not present in additive contexts. This serves as a reminder that a comprehensive unbiased understanding of the acquisition of L2 prosody cannot be achieved unless it incorporates findings based on subtractive bilingualism.

21.3 L2 Sentence Intonation

21.3.1 Sources of Variation

Sentence intonation has become a fertile ground for systematic study thanks to theories like autosegmental phonology (e.g., Ladd, 2008) and transcription systems like Tones and Break Indices (ToBI; Beckman, Hirschberg, & Shattuck-Hufnagel, 2006 and references therein). Following the line taken by autosegmental phonology (Goldsmith, 1976), segments and tones are defined as autosegments generated in different tiers, with phones in the segmental tier and tones in the melodic tier, which align with each other in accordance with general principles. While the complete set of possible tones and phonemes is universal, each language uses a particular subset of them. These principles were applied to study sentence intonation first in English and Japanese (Beckman et al., 2006) but soon across many other languages, generating language-specific prosodic grammars (e.g., Spanish ToBI: Beckman et al., 2002; Mandarin ToBI: Peng et al., 2005), which were further adapted to express dialectal variation in both L1 (Clopper & Smiljanic, 2011; Cruz, 2013, Gili Fivela et al., 2015; Prieto & Roseano, 2010) and L2 varieties (see, e.g., Mennen's [2015] L2 Intonation Learning theory or LILt). At the same time, these variety-specific prosodic grammars have advanced our understanding of prosodic typology by showing how varieties can be grouped according to their intonational structure (Jun, 2014). Recently, ToBI has also been used to analyze pathological speech (e.g., Frota et al., 2021).

Simultaneously, the growing field of sociophonetics has begun to incorporate prosody into research on social meaning (Holliday & Villarreal, 2020). In short, over the past few decades intonational theories and transcription systems have proven to be of great utility in exploring the different sources of variation that affect sentence intonation. In this section, two such sources of variation will be reviewed, L1–L2 typological distance with regard to the expression of narrow focus (Section 21.3.2), and the social aspects of intonation (Section 21.3.3).

21.3.2 Prosody-Based Generalizations About the Expression of Focus

Typologically distant languages tend to use different linguistic strategies to express equivalent meanings, as is the case, for example, with English and Spanish in their expression of narrow focus (e.g., Zubizarreta, 1998). This is illustrated in short dialogues (21.1)–(21.3), where “Mary/María,” “bone/hueso,” and “renting/alquilando” constitute the information requested in each question, respectively. The narrow focus applied to these words is expressed prosodically by marking the word as the most prominent in the sentence, that is, it bears the nuclear stress (NS) of the utterance (indicated with capitalized letters in the examples). In English, NS is assigned by a nuclear stress rule (NSR) which mandates not only the assignment of prominence to the word in focus (e.g., MARY) but also the de-accenting of post-focal words (e.g., “came” in (21.1)). Phonetically, prominence on “Mary,” “bone,” and “renting” is signaled in the speech signal by enlarging the pitch range, duration, and intensity of the stressed syllable. Post-focal clause de-accenting is expressed by compressing pitch range, duration, and intensity (Cole et al., 2019; Ladd, 2008). By contrast, according to the NSR applied to Spanish, the word with prominence must be positioned at the end of the intonation phrase (e.g., *la está ALQUILANDO* in (21.3)). Consequently, the phonetic strategy of enlarging the suprasegmental cues of the in-focus word is relevant to the English NSR only because prominence in the Spanish NSR strategy is expressed exclusively by sentence position.

(21.1) English: A: Who came yesterday? B: MARY did.

Spanish: A: ¿Quién vino ayer? B: Vino MARÍA.

(21.2) English: A: What’s the dog chewing? B: It’s chewing a BONE.

Spanish: A: ¿Qué está mordiendo el perro? B: Está mordiendo un HUESO.

(21.3) English: A: Did John buy that green house? B: No, John is RENTING it.

Spanish: A: ¿Se compró Juan la casa verde? B: No, la está ALQUILANDO.

Research on the acquisition of focus in L2 English has described, on the one hand, the difficulty consistently experienced by Spanish speakers as they unlearn the Spanish NSR to acquire the English NSR (Ortega-Llebaria & Colantoni, 2014; Zubizarreta & Nava, 2011). This difficulty manifests itself in the following patterns. First, Spanish speakers start out by applying phonetic enhancement at the end of L2 English narrow focus statements, but then, as their proficiency increases, they begin to move it to other sentence positions, suggesting an awareness that they must apply a different NSR. However, after acquiring the English NSR to express focus, they consistently fail to generalize this strategy to thematic and categorical meanings. Finally, they have great a deal of difficulty learning to de-accent the post-focal clause, so, instead of de-accenting, they use repair strategies such as the insertion of a pause after the subject focus (e.g., MARY || came).

On the other hand, studies of L2 English learners have shown that speakers of Beijing Mandarin – in whose L1 the same NSR applies as in English – consistently obtain more native-like scores than Spanish speakers in all perception and production tasks (Ortega-Llebaria & Colantoni, 2014; Xu, Chen, & Wang, 2012). That said, it should be noted that although all Mandarin Chinese dialects use the same NSR to express narrow focus, these dialects differ in their use of post-focal compression, such that the Mandarin spoken in Beijing has post-focal compression while the Mandarin spoken in Taiwan does not (Xu et al., 2012). As a result, only Beijing speakers use post-focal compression as a cue to the perception of narrow focus in their own dialect. Interestingly, this asymmetry in the perception of post-focal compression has been seen to transfer to the perception of L2 English narrow focus, resulting in L1 Beijing Mandarin speakers obtaining higher scores than their Taiwanese counterparts (Adams & Hellmuth, 2022).

In summary, L1–L2 typological distance constitutes a key tool for interpreting L2 focus acquisition by speakers of different L1s. Differences between the NSRs applying in an L1 and an L2 are likely to impede the acquisition of prosody, as data from Spanish speakers of English illustrate. Moreover, if their own L1 does not use post-focal compression, learners experience difficulties producing and perceiving post-focal compression in an L2, as illustrated by the L2 English of native speakers of the Beijing and Taiwan Mandarin Chinese varieties.

21.3.3 Social Aspects of Intonation: Accommodation in Subtractive Contexts

Accommodation refers to the changes in language style that a speaker effectuates to gain social acceptability from an in-group and/or out-group, leading to convergence between language styles, divergence, or maintenance. In principle, accommodation differs from transfer in that the intention underlying the former is to gain social acceptability. In other words, in accommodation the use of a language structure is linked to a social meaning, whereas in transfer it is simply the result of applying L1 parameters to an L2. In practice, this fine line

between transfer and accommodation often becomes blurred, so that, for example, although a foreign-accent trait in an immigrant's speech is initially the result of L1-to-L2 transfer, it can be assigned a social meaning by the host L2-speaking community, which later the speaker may use intentionally to signal convergence with their own immigrant in-group.

Accommodation in sentence intonation has been observed in speakers' choice of pitch accents and in their use of pitch range. The dynamics of this kind of accommodation can be seen in Spain, for example, in the pitch accents used in absolute interrogatives by bilingual speakers from the island of Majorca. In Standard Peninsular Spanish, absolute interrogatives are produced with an ascending F0 intonation transcribed as L*H%. However, in his analysis of the speech of Catalan-Spanish balanced bilinguals growing up in Majorca, Simonet (2008) found a general tendency to use the Majorcan Catalan HL* L% descending contour in absolute interrogatives when they spoke Spanish, indicating a convergence toward Catalan which was modulated by the age and language dominance of the bilingual speakers. The Majorcan Catalan descending contour occurred most among the younger Catalan-dominant speakers, Catalan-dominant bilinguals, then the younger Spanish-dominant bilinguals. By contrast, the Spanish ascending L*H% predominated among the older Spanish-dominant bilinguals. These patterns of use clearly illustrated the bidirectionality of the convergence effects, that is, from L1-to-L2 in Catalan-dominant speakers and from L2-to-L1 in Spanish-dominant speakers. However, it was not clear for the Spanish-dominant group whether the use of the Catalan HL* L% in their L1 Spanish came from their L2 Catalan – which would indicate direct accommodation – or from the Spanish variety spoken in Majorca – which would indicate indirect accommodation. More recently, Romera and Elordieta (2013) provided evidence from the same context of indirect accommodation in the speech of monolingual Spanish-speaking immigrants in Majorca by showing that they produced the descending HL*L% in their L1 Spanish interrogatives after living in Majorca for less than four years, despite not having learned any Catalan during that time. Interestingly, that accommodation was modulated by affective variables such as the degree of interest on the part of the speaker in becoming integrated in the host community.

Further evidence – also from Spain – of the role of affective factors in accommodation comes from the patterns detected in the speech of bilingual speakers of Lekitio Basque and Spanish by Elordieta and Romera (Elordieta & Romera 2021; Romera & Elordieta 2020). Like in Majorcan Catalan, absolute interrogatives exhibited descending intonation, which could also appear in the variety of Spanish spoken in the Basque Country when the speakers were accommodating. In these cases, affective variables could even override social factors such as age or gender and linguistic factors such as language dominance. Studies of this sort make it clear that any analysis of prosodic accommodation must include not just social and linguistic but also affective variables.

Pitch range has also been shown to be susceptible to accommodation. Human societies have historically projected the obvious link between vocal pitch and gender onto issues of power, such that higher pitch is related to femininity and submissive attitudes, and lower pitch to masculinity and dominance (Gussenhoven, 2004; Ohala, 1984; Winter et al., 2021). Not surprisingly, the few studies that have looked at how bilingual speakers accommodate to the pitch range of one or the other language have shown that gender plays an important role. Ordin and Mennen (2015, 2017), for example, examined the speech of Welsh-English bilinguals. In general, Welsh has a larger pitch range than English, which is achieved by means of the higher F0 maxima of pitch accents. The authors found that, interestingly, most male participants in the study did not change pitch ranges when switching between languages, failing to accommodate presumably because the higher pitch range sounded “feminine,” while females consistently did so, showing a distinct pitch range for each language. These differences were attributed by the authors to the sociocultural factors linking pitch and gender on the one hand and power on the other.

In a similar study, de Leeuw (2020) examined the pitch level and pitch range of Germans living in Anglophone Canada. In general, when they were speaking German, the Germans’ natural pitch range was intermediate between that of Canadian English-speaking females and that of Canadian English-speaking males. As expected, the female Germans converged with female Canadian speech by increasing and widening their F0 range in both English and German. Unexpectedly, however, the German males increased their pitch range, more closely approximating the wider pitch range of Canadian females than the narrower range of Canadian males. As a result, German males’ English diverged from the Canadian males’ English. The authors postulated that this divergence reflected an attempt by the male Germans to exploit the aforementioned associations between pitch and power to distance themselves from the aggressive stereotypes of Germans that they knew were held in Canada.

Add this reference: Pešková, A. (2023). L2 Spanish and Italian intonation. BoD—Books on Demand.

Recent research (Mennen et al., 2022) has applied the LILt model (Mennen, 2015) to investigate the L1 prosody of long-term English-speaking expatriates in Austria. The results revealed that L2-to-L1 prosodic accommodation affected the four dimensions outlined in the model, to wit, the phonological dimension or pitch accent inventory, the phonetic dimension or pitch accent implementation, the semantic dimension or form-meaning mapping, and pitch accent frequency of use, illustrating the great permeability of L1 prosody. Because some L2-to-L1 patterns could not be accounted for by purely linguistic concepts such as markedness, these results provided further proof of the need to add social and attitudinal factors to linguistic variables when analyzing the acquisition of prosody by bilinguals.

21.3.4 Summary

Intonation theories and transcription systems have generated a wealth of language-specific prosodic grammars, in the process uncovering multiple

factors that affect variation in prosody and triggering the use of prosodic typologies. Both issues have had a profound effect on the study of L2 sentence intonation, giving rise, for example, to the LILt model of L2 prosody (Mennen, 2015). As for prosodic typologies, research has shown that typological distance between an L1 and an L2 has an impact on learner abilities to approximate the expression of focus by monolingual native speakers of the L2, and just as it does for L2 stress and rhythm (Section 21.4). Social meaning is also an important source of variation in sentence intonation. Accommodation literature in particular has uncovered evidence of the effect of affective variables on L2 sentence intonation, which can at times be stronger than the effect of more obvious linguistic and social factors, especially in multilingual and multicultural communities.

21.4 L2 Rhythm

21.4.1 Definition and Contextualization

In general, rhythm refers to the sensation of isochrony conveyed by a repetition pattern in auditory stimuli, most obviously exemplified by the 1–2–3 1–2–3 beat of a waltz. In speech, the perceived isochrony shown by primary-stresses, syllables, and morae has been used to classify languages into stress-timed, syllable-timed, and morae-timed rhythmic classes. Although production studies have failed to find acoustic correlates to these rhythmic classes (see Arvaniti [2012] for a historical overview), supporting evidence has come from adult and infant perceptual research (e.g., Murty, Otake, & Cutler, 2007; Nazzi, Bertoncini, & Mehler, 1998). In one study, for example, Ramus and colleagues measured the vowel and consonant interval durations of the sentence stimuli employed in previous infant discrimination experiments and found that the percentage of vocalic intervals in those sentences (%V) together with variations in the duration of vowel and consonant intervals (ΔV and ΔC) captured rhythm classes (Ramus, Nespor, & Mehler, 1999; see Nazzi & Ramus [2003] for a later review). Ramus' duration-based rhythm measures were refined in subsequent research. On the other hand, strong criticism questioning the suitability of using duration-based measures to distinguish rhythmic classes (Arvaniti, 2012) led to research on new measures based on acoustic dimensions other than duration. Advances in L2 rhythm research have proven to be contingent on the fine-tuning of these new measures.

21.4.2 Measures

To refine Ramus' duration-based measures, researchers proposed the following modifications. Frota and Vigário's (2001) $\Delta\%C$ and $\Delta\%V$ measured the variability of interval duration relative to sentence duration. Grabe and Low's (2002) Pairwise Variability Index (PVI) measured duration variation between

adjacent intervals in a sentence, capturing the fact that the perceived duration of an interval is relative to the duration of the surrounding intervals. Dellwo's (e.g., 2010) Varco normalized interval durations for speech rate. Applying PVI and Varco for the first time to L2, White and Mattys (2007) showed that these measures were sensitive to capturing differences between L1 and L2 pairs, such as L1 Spanish compared with the L2 Spanish of L1 English speakers. In the following decade, the original PVI and Varco measures were subjected to further refinements in the light of their sensitivity to various factors such as L2 proficiency levels (e.g., Kawase, Kim, & Davis, 2016; Ozaki, Yazawa, & Kondo, 2017), the differences detected between large speech corpora of the same language being spoken as an L1 and an L2 (e.g., Lai, Evanini & Zechner, 2013), human judgments of rhythmicity in L2 data (Kinoshita & Sheppard, 2011), and dialectal variation (e.g., Cruz, 2013).

Despite these improvements, the notion that the various rhythmic classes could be accurately defined using duration-based measures came under sharp attack (Arvaniti, 2009, 2012; Barry, Andreeva, & Koreman, 2009), representing a turning point in rhythm research. While Arvaniti (2012) questioned the validity of duration-based rhythmic classes, Gasparini et al. (2021) proposed a shift in focus from rhythmic classes to metrics proper based on their meta-analysis of forty-two infant studies. They showed that rhythmic classes failed to account for infants' language discrimination abilities better than successive consonant-duration interval variability. At about the same time, neuroscience research revealed the importance of synchronizing time windows of different sizes when describing isochrony (e.g., Poeppel & Assaneo, 2020), showing the need to add larger timescales such as syllables and words to the vowel and consonant intervals of duration metrics and thus underscoring the need to include pitch-based and intensity-based measures in any analysis of prosody.

With regard to pitch-based measures, Polyanskaya, Busà, and Ordin (2020) were the first to explore the F0 contribution to the sensation of isochrony cross-linguistically. Based on Jun's (2014) concept of macro-rhythm, these authors applied to Italian and English pitch-based measures such as the distance between F0 peaks and F0 valleys, the magnitude of pitch excursions, and the number of tonal targets per intonational unit. The tonal rhythm of Italian proved to be more regular than that of English, lending support to Jun's predictions. These results have opened a door to the investigation of F0-based rhythm measures in L2s and their relationship with duration-based measures

(e.g., Nagao & Ortega-Llebaria, 2021; Ortega-Llebaria, Silva & Nagao, 2023)

With regard to intensity-based measures, many have been proposed. He (2012) applied to intensity the PVI and Varco measures previously used on duration. These intensity measures accounted for the perceived rhythmic differences between L1 English and Mandarin-accented English that duration-based measures failed to capture. Quené et al. (2014) operationalized rhythm as peak frequency in the intensity envelope spectrum, normalized to syllables per second. This measure captured rhythmic convergence between L2 English and that of the monolingual community. Of particular interest are the recent

amplitude envelope-based measures of rhythm because they reflect overlapping timescales of different sizes, thus overcoming the limitation of a single timescale displayed by previous duration-based measures. Pellegrino, Schwab, and Dellwo (2021) directly compared the effects of amplitude envelope and segment duration on rhythm while controlling for the speech rate of native Italian and L2 Italian as spoken by L1 German speakers. The effect of amplitude envelope in the perception of rhythmic differences between L1 and L2 Italian remained once the effect of segment duration was controlled for. Additional evidence from Frota et al. (2022) showed that wideband amplitude envelopes of different bands captured rhythmic differences between Brazilian Portuguese, European Portuguese, and German.

Besides these acoustic correlates, gestures have also been investigated because of their proven synchronization with prosody. For example, Rathcke et al. (2021) asked native English speakers to reproduce the beat of a looped sentence they had heard by tapping their fingers and found that the participants reproduced the rhythmic structure of sentences with a consistent anchoring of the taps on vowel onsets. Tying this to L2 acquisition research, Prieto and colleagues showed that the addition of beat-gestures (up-down hand movements) to L2 rhythm perception training significantly improved learners' pronunciation and comprehension of the language (e.g., Baills, 2022; Gluhareva & Prieto, 2017).

21.4.3 L2 Rhythm in Additive Bilingual Contexts

Earlier research using duration-based measures not only yielded inconsistent results with regard to L2 proficiency effects (e.g., Guilbault, 2002) and L2 length of exposure (e.g., Gut, 2009) but also exposed the difficulty of interpreting these measures. For instance, although L1 Japanese speakers increased duration variability in their L2 English, and these measures progressively approached English monolingual norms, they did not reflect the typical duration patterns of English stress (Grenon & White, 2008). Crucially, Arvaniti (2012) identified speaker variation, elicitation method, and syllable complexity as variables that all had larger effects than rhythmic class. Accordingly, various authors designed experiments in which these effects were controlled for (Arvaniti, 2009, 2012; Gut, 2012; Prieto et al., 2012), resulting in more findings that were consistent.

These new results indicate that L2 duration-based isochrony is learnable and that L2 proficiency modulates this learning (e.g., da Silva & Barbosa, 2019; Gut, 2012). For example, Ordin and Polyanskaya (2015) and Li and Post (2014) showed that L1 German, L1 French, and L1 Mandarin learners of L2 English progressively improved their L2 duration-based patterns along with their English proficiency. However, in both studies German speakers showed more overall improvement than French or Mandarin speakers, illustrating the effect of L1–L2 typological distance. Directionality was also a modulating factor. Van Maastricht et al. (2019) showed that while both Dutch learners of

Spanish and Spanish learners of Dutch improved L2 rhythm, Dutch learners of Spanish achieved higher scores. Because L1–L2 typological distance was the same, the authors attributed this learning difference to the respective phonological grammars of the two languages. Two other factors that seem to modulate L2 learning are musical abilities (Cason et al., 2020) and gestures (e.g., Gluhareva & Prieto, 2017).

In addition to these factors, a universal learning pattern has been identified whereby, when learning a stress-timed L2, learners will initially employ syllable-timed patterns, regardless of the rhythm class of their L1. For example, exactly like French or Mandarin speakers, German speakers start out speaking L2 English using a more syllable-timed pattern, despite the fact that German, unlike French or Mandarin, is not a syllable-timed language (Ordin & Polyanskaya, 2015; Li & Post, 2014).

21.4.4 L2 Rhythm in Subtractive Bilingual Contexts

Contexts of nonadditive bilingualism provide specific learning situations that modulate the factors affecting L2 acquisition in particular ways. It is in these contexts that bidirectional transfer between L1 and L2 has been observed most frequently. On the one hand, there is abundant evidence of L1-to-L2 transfer even after the L1 is no longer spoken in the community (Benet et al., 2012; Gabriel & Kireva, 2014). For example, *Porteño* Spanish, the Spanish spoken nowadays in Buenos Aires, shows traits of Italian prosody due to the prolonged period of Italian-Spanish language contact resulting from the mass migration of Italians to the city between 1860 and the early twentieth century (Colantoni & Gurlekian, 2004). This was demonstrated by Benet, Gabriel, and colleagues (Benet et al., 2012; Gabriel & Kireva, 2014), who compared the rhythmic patterns of *Porteño* Spanish with those of Castilian Spanish, Italian, and L2 Spanish spoken by Italian speakers and found that *Porteño* Spanish displayed values that were closer to L1 Italian and L2 Spanish spoken by Italian speakers than to Castilian Spanish.

On the other hand, L2-to-L1 transfer has been frequently reported in bilingual communities where the L2 becomes the dominant language but does not displace the L1. Coetzee et al. (2015) studied bilingual speakers of Afrikaans and Spanish living in Argentinian Patagonia. After decades of speaking only Afrikaans (a stress-timed language), in the second half of the twentieth century Spanish (a syllable-timed language) gradually became the dominant language of the community. At the time of the study, the rhythmic patterns of the bilingual speakers' Spanish were identical to those of monolingual controls. However, the rhythmic patterns of the bilinguals' Afrikaans showed L2 transfer from Spanish. Similarly, children in Botswana are raised bilingual in Setswana (a syllable-timed language) and English (a stress-timed language) within a diglossic context where the colonial language, English, is the prestige variety used in public settings (Bagwasi, 2021). Schools can use either English or Setswana as their vehicular language. In schools where

English is the vehicular language, the use of Setswana is relegated to family settings and English eventually becomes the pupils' dominant language, with the result that the children transfer the rhythmic patterns of their L2, English, to their L1, Setswana (Sebina, Setter, & Daller, 2021).

21.4.5 Summary

After controlling for elicitation methods and the syllable-structure of test stimuli (e.g., Arvaniti, 2009, 2012), duration-based Varco and PVI measures of L2 output yield consistent results which reveal that the factors of L2 proficiency, L1–L2 typological distance, directionality, musical abilities, and gestures modulate the learning of L2 duration-based isochrony. In addition, a universal L2 learning pattern has been identified such that learners of stress-timed L2s start out by producing syllable-timed patterns independently of their L1. Many intensity-based measures of isochrony have been shown to be effective for capturing L2 learning patterns. Further, F0-based measures of isochrony are just starting to be used in L2 research. And L2 learning in subtractive contexts where the L2 becomes the dominant language in a community (e.g., bilingual Afrikaans–Spanish-speaking communities in Argentina, bilingual Setswana–English-speaking communities in Botswana) has shown both L1-to-L2 and L2-to-L1 transfer (Section 21.3.3). Future research needs to disentangle the role of social, linguistic, and cognitive factors in these contexts.

21.5 Conclusions and Directions for Future Research

Research from the last twenty-five years demonstrates that adults are capable of learning L2 stress, L2 duration-based rhythm, and L2 sentence intonation. Two findings are of particular relevance. First, various factors affecting L2 prosody have emerged, mostly in contexts of subtractive bilingualism. Whereas L1-to-L2 transfer is ubiquitous, L2-to-L1 transfer can take place in situations of language contact where the L2 is or becomes the dominant language, as in the case of L1 Afrikaans–L2 Spanish speakers in Patagonia (Coetzee et al., 2015) or L1 Dutch speakers who are long-term residents of Anglophone Canada (Bruggeman, 2016). This L2-to-L1 transfer has been observed in L2 stress, sentence intonation, and rhythm, as well as in the four dimensions outlined in LILt (Mennen, 2015). While in contexts of additive bilingualism, speakers' perception and production of a prosodic feature are equally high or equally low, an inconsistent high perception but low production of Spanish stress has been observed in heritage Spanish speakers residing in the USA (Kim, 2020). This pattern has been linked to the situation of subtractive bilingualism: the knowledge of Spanish acquired by heritage speakers through early exposure to Spanish at home can be lost if this use is interrupted early in development (e.g., Rao, 2019). Prosodic accommodation

in multilingual communities conveys social meaning and is mainly modulated by affective and social factors rather than by factors traditionally emphasized in second language acquisition like L2 proficiency and language dominance (Elordieta & Romera, 2021). Altogether, this cumulative evidence underlines the fact that to date most research into the acquisition of L2 prosody has been carried out in contexts of additive bilingualism, and considerable work from contexts of subtractive bilingualism must be done to correct this imbalance.

The second main research finding relates to the central role played by form-meaning mapping in explaining speakers' ability to approximate L2 prosodic targets. With regard to the acquisition of L2 stress, sentence intonation, or rhythm, speakers approximate an L2 to a lesser degree when learning the L2 either requires remapping L1 suprasegmental cues to new L2 meanings, such as when French speakers map the demarcative function of suprasegmental cues in their L1 onto the culminative function of stress in Spanish, or requires speakers to express the same meaning with prosodic features in one language and nonprosodic features in the other, as can be seen when English speakers learn Spanish syntactic-based focus. In both cases, L2 learning implies remapping prosodic forms to new meanings. Conversely, speakers approximate L2 targets more closely when no remapping is involved, that is, when learning an L2 prosodic target requires only transferring L1 suprasegmental cues to the L2 to express similar meanings.

These two main findings suggest the following routes for future research. First, to overcome bias favoring additive bilingualism, L2 prosody research needs to include more situations of subtractive bilingualism such as heritage speakers, long-term migrants, and pidgins and creoles, to name but a few. Reducing this bias together with taking an interdisciplinary approach that combines linguistic, social, and affective variables will help to unravel the reasons why specific factors modulating L2 prosody emerge only in certain situations of language contact. In short, it will help us to understand why reactions to "Gravy?" spoken with Indian English intonation can differ so sharply in the contexts of a British airport cafeteria or an Indian university cafeteria, as we saw in Section 21.1.

Second, the central role of form-meaning mappings in L2 prosody highlights the need to further our understanding of the meanings that prosody conveys. However, L2 prosody research has emphasized forms over meanings, and consequently only meanings relevant to the particular prosodic forms under study have been investigated, such as the culminative versus demarcative functions of L2 stress or the social meanings related to accommodation patterns. With the exception of a few recent papers (Barbosa & Niebuhr, 2020; Orrico & D'Imperio, 2020; Rao, Ye, & Butera, 2022; Taguchi, Hirsch, & Kang, 2022), the prosody of pragmatic meanings such as evidentiality and of emotional meanings such as those characterizing charismatic speech is mostly being studied in connection with L1s (e.g., Escandell & Prieto, 2020; Niebuhr, Thumm, & Michalsky, 2018), with research linked to L2s being still relatively meager (for a review see Kang & Kermad, 2019).

Finally, L2 prosody research has thus far mainly explored domains up to the sentence, bypassing discourse. To fill this gap, it is crucial to incorporate work that focuses on discourse pragmatics and conversation models (Ward, 2019). Studying discourse requires big data and efficient (semi-)automated L2 prosody annotation methodologies (Kang, Johnson & Kermad, 2021), which in turn involves interdisciplinary research among phoneticians, semanticists, pragmatists, sociolinguists, and computational linguists.

References

- Adams, J. & Hellmuth, S. (2022). Taiwanese and Beijing Mandarin listeners' perception of English focus prosody. In *Proceedings of Speech Prosody 2022*, pp. 744–748. <https://doi.org/10.21437/SpeechProsody.2022-151>.
- Anyu, U. (2021). Critical race pedagogy for more effective and inclusive world language teaching. *Applied Linguistics*, 42(6), 1055–1069.
- Armstrong, M., Breen, M., Gooden, S., Levon, E., & Yu, K. M., eds. (2022). Sociolectal and dialectal variation in prosody [special issue]. *Language and Speech*, 65(4), 783–790.
- Arvaniti, A. (2009). Rhythm, timing and the timing of rhythm. *Phonetica*, 66(1–2), 46–63.
- Arvaniti, A. (2012). The usefulness of metrics in the quantification of speech rhythm. *Journal of Phonetics*, 40(3), 351–373.
- Bagwasi, M. M. (2021). Education, multilingualism and bilingualism in Botswana. *International Journal of the Sociology of Language*, 2021(267–268), 43–54.
- Baills, F. (2022). Using the hands to embody prosody boosts phonological learning in a foreign language. [Doctoral dissertation, Universitat Pompeu Fabra].
- Barbosa, P. & Niebuhr, O. (2020). Persuasive speech is a matter of acoustics and chest breathing only. In M. Elmentaler & O. Niebuhr, eds., *An den Rändern der Sprache*. Frankfurt: Peter Lang, pp. 551–578.
- Barry, W., Andreeva, B., & Koreman, J. (2009). Do rhythm measures reflect perceived rhythm? *Phonetica*, 66(1–2), 78–94.
- Beckman, M. E., Díaz-Campos, M., McGory, J. T., & Morgan, T. A. (2002). Intonation across Spanish, in the Tones and Break Indices Framework. *Probus*, 14(1), 9–36.
- Beckman, M. E., Hirschberg, J., & Shattuck-Hufnagel, S. (2006). The original ToBI system and the evolution of the ToBI system. In S.-A. Jun, ed., *Prosodic Typology: The Phonology of Intonation and Phrasing*. Oxford: Oxford University Press, pp. 9–54.
- Benet, A., Gabriel, C., Kireva, E., & Pešková, A. (2012). Prosodic transfer from Italian to Spanish: Rhythmic properties of L2 speech and Argentinean Porteño. In Q. Ma, H. Ding, & D. Hirst, eds., *Speech Prosody 2012*. Shanghai: Tongji University Press, pp. 438–441.
- Bruggeman, L. (2016). Native, dominance, and the flexibility of listening to spoken language. [Doctoral dissertation, Western Sydney University].
- Bruggeman, L., Yu, J., & Cutler, A. (2022). Listener adjustment of stress cue use to fit language vocabulary structure. In *Proceedings of Speech Prosody 2022*, pp. 264–267. <https://doi.org/10.21437/SpeechProsody.2022-54>.

- Cason, N., Marmursztejn, M., D'Imperio, M., & Schön, D. (2020). Rhythmic abilities correlate with L2 prosody imitation abilities in typologically different languages. *Language and Speech*, 63(1), 149–165.
- Choi, W. (2022). Theorizing positive transfer in cross-linguistic speech perception: The acoustical-attentional-contextual hypothesis. *Journal of Phonetics*, 91, 101–135.
- Choi, W., Tong, X., & Samuel, A. G. (2019). Better than native: Tone language experience enhances second language English lexical stress discrimination in Cantonese-English bilinguals. *Cognition*, 189, 188–192.
- Clopper, C. G. & Smiljanic, R. (2011). Effects of gender and regional dialect on prosodic patterns in American English. *Journal of Phonetics*, 39(2), 237–245.
- Coetzee, A. W., García-Amaya, L., Henriksen, N., & Wissing, D. (2015). Bilingual speech rhythm: Spanish-Afrikaans in Patagonia. In M. Wolters, J. Livingstone, B. Beattie, R. Smith, M. MacMahon, J. Stuart-Smith, & J. M. Scobbie, eds., *Proceedings of the 18th International Congress of Phonetic Science (ICPhS 2015)*. Glasgow: Glasgow University. www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2015/Papers/ICPHS0911.pdf.
- Colantoni, L. & Gurlekian, J. (2004). Convergence and intonation: Historical evidence from Buenos Aires Spanish. *Bilingualism: Language and Cognition*, 7(2), 107–119.
- Cole, J., Hualde, J. I., Smith, C. L., et al. (2019). Sound, structure and meaning: The bases of prominence ratings in English, French and Spanish. *Journal of Phonetics*, 75, 113–147.
- Cooper, N., Cutler, A., & Wales, R. (2002). Constraints of lexical stress on lexical access in English: Evidence from native and non-native listeners. *Language and Speech*, 45(3), 207–228.
- Cruz, M. (2013). Prosodic variation in EP: Phrasing, intonation and rhythm in Central-Southern varieties. [Doctoral dissertation, University of Lisbon].
- Cutler, A. (2005). Lexical stress. In D. B. Pisoni & R. E. Remez, eds., *The Handbook of Speech Perception*. Oxford: Blackwell, pp. 264–289.
- Cutler, A. (2012). *Native Listening: Language Experience and the Recognition of Spoken Words*. Cambridge, MA: MIT Press.
- Cutler, A. & Jesse, A. (2021). Word stress in speech perception. In J. S. Pardo, L. C. Nygaard, R. E. Remez, & D. B. Pisoni, eds., *The Handbook of Speech Perception*. Hoboken, NJ: Wiley, pp. 239–265.
- Da Silva Junior, L. J. & Barbosa, P. A. (2019). Speech rhythm of English as L2: An investigation of prosodic variables on the production of Brazilian Portuguese speakers. *Journal of Speech Sciences*, 8(2), 37–57.
- De Leeuw, E. (2020). The Frequency Code and gendered attrition and acquisition in the German-English heritage language community in Vancouver, Canada. In B. Brehmer & J. Treffers-Daller, eds., *Lost in Transmission: The Role of Attrition and Input in Heritage Language Development*. Amsterdam: John Benjamins, pp. 230–253.
- Dellwo, V. (2010). Influences of speech rate on the acoustic correlates of speech rhythm: An experimental phonetic study based on acoustic and perceptual evidence. [Doctoral dissertation, Bonn University].
- Dupoux, E., Pallier, C., Sebastian, N., & Mehler, J. (1997). A destressing “deafness” in French? *Journal of Memory and Language*, 36(3), 406–421.
- Dupoux, E., Peperkamp, S., & Sebastián-Gallés, N. (2001). A robust method to study stress “deafness.” *Journal of the Acoustical Society of America*, 110(3), 1606–1618.

- Dupoux, E., Peperkamp, S., & Sebastián-Gallés, N. (2010). Limits on bilingualism revisited: Stress “deafness” in simultaneous French-Spanish bilinguals. *Cognition*, 114(2), 266–275.
- Elordieta, G. & Romera, M. (2021). The influence of social factors on the prosody of Spanish in contact with Basque. *International Journal of Bilingualism*, 25(1), 286–317.
- Escandell-Vidal, V. & Prieto, P. (2020). Pragmatics and prosody in research on Spanish. In D. A. Koike & J. C. Félix-Brasdefer, eds., *The Routledge Handbook of Spanish Pragmatics*. New York: Routledge, pp. 149–166.
- Flores, N. & García, E. S. (2020). Power, language, and bilingual learners. In N. S. Nasir, C. D. Lee, R. Pea, & M. McKinney de Royston, eds., *Handbook of the Cultural Foundations of Learning*. New York: Routledge, pp. 178–192.
- Frota, S., Cruz, M., Cardoso, R., et al. (2021). (Dys)Prosody in Parkinson’s disease: Effects of medication and disease duration on intonation and prosodic phrasing. *Brain Sciences*, 11(8), 1100.
- Frota, S. & Vigário, M. (2001). On the correlates of rhythmic distinctions: The European/Brazilian Portuguese case. *Probus*, 13, 247–275.
- Frota, S., Vigário, M., Cruz, M., Hohl, F., & Braun, B. (2022). Amplitude envelope modulations across languages reflect prosody. In *Proceedings of the 11th International Conference on Speech Prosody 2022*, pp. 688–692. <http://doi.org/10.21437/SpeechProsody.2022-140>.
- Gabriel, C. & Kireva, E. (2014). Prosodic transfer in learner and contact varieties: Speech rhythm and intonation of Buenos Aires Spanish and L2 Castilian Spanish produced by Italian native speakers. *Studies in Second Language Acquisition*, 36(2), 257–281.
- Gasparini, L., Langus, A., Tsuji, S., & Boll-Avetisyan, N. (2021). Quantifying the role of rhythm in infants’ language discrimination abilities: A meta-analysis. *Cognition*, 213, 104757.
- Gili Fivela, B., Avesani, C., Barone, M., et al. (2015). Intonational phonology of the regional varieties of Italian. In S. Frota & P. Prieto, eds., *Intonation in Romance*. Oxford: Oxford University Press, pp. 140–197.
- Gluhareva, D. & Prieto, P. (2017). Training with rhythmic beat gestures benefits L2 pronunciation in discourse-demanding situations. *Language Teaching Research*, 21(5), 609–631.
- Goldsmith, J. (1976). Autosegmental phonology. [Doctoral dissertation, Massachusetts Institute of Technology].
- Grabe, E. & Low, E. L. (2002). Durational variability in speech and the rhythm class hypothesis. *Papers in Laboratory Phonology*, 7(1982), 515–546.
- Grenon, I. & White, L. (2008). Acquiring rhythm: A comparison of L1 and L2 speakers of Canadian English and Japanese. In H. Chan, H. Jacob, & E. Kipia, eds., *Proceedings of the 32nd Boston University Conference on Language Development*. Somerville, MA: Cascadia Press, pp. 155–166.
- Guilbault, C. P. G. (2002). The acquisition of French rhythm by English second language learners. [Doctoral dissertation, University of Alberta, Edmonton].
- Gumperz, J. J. (1997). Interethnic communication. In N. Coupland & A. Jaworski, eds., *Sociolinguistics*. London: Palgrave, pp. 395–407.
- Gussenhoven, C. (2004). *The Phonology of Tone and Intonation*. Cambridge: Cambridge University Press.

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Gut, U. (2009). *Non-native speech: A Corpus-Based Analysis of Phonological and Phonetic Properties of L2 English and German*. Frankfurt: Peter Lang.

Gut, U. (2012). Rhythm in L2 speech. *Speech and Language Technology*, 14(15), 83–94.

He, L. (2012). Syllabic intensity variations as quantification of speech rhythm: Evidence from both L1 and L2. In *Proceedings of 6th International Conference on Speech Prosody 2022*, 466–469. <https://doi.org/10.5167/uzh-127263>.

Holliday, N. & Villarreal, D. (2020). Intonational variation and incrementality in listener judgments of ethnicity. *Laboratory Phonology: Journal of the Association for Laboratory Phonology*, 11(1), 3.

Jun, S.-A. (2014). Prosodic typology: By prominence type, word prosody, and macro-rhythm. In S.-A. Jun, ed., *Prosodic Typology II: The Phonology of Intonation and Phrasing*. Oxford: Oxford University Press, pp. 520–539.

Kachru, B. B. (1992). Teaching world Englishes. *The Other Tongue: English Across Cultures*, 2(2), 355–365.

Kang, O. & Kermad, A. (2019). Prosody in L2 pragmatics research. In N. Taguchi, ed., *The Routledge Handbook of Second Language Acquisition and Pragmatics*. New York: Routledge, pp. 78–92.

Kang, O., Johnson, D. O., & Kermad, A. (2021). *Second Language Prosody and Computer Modeling*. New York: Routledge.

Kawase, S., Kim, J., & Davis, C. (2016). The influence of second language experience on Japanese-accented English rhythm. In *Proceedings of Speech Prosody 2016*, pp. 746–750. <https://doi.org/10.21437/SpeechProsody.2016-153>.

Kim, H. & Tremblay, A. (2021). Korean listeners' processing of suprasegmental lexical contrasts in Korean and English: A cue-based transfer approach. *Journal of Phonetics*, 87, 101059.

Kim, J. Y. (2020). Discrepancy between heritage speakers' use of suprasegmental cues in the perception and production of Spanish lexical stress. *Bilingualism: Language and Cognition*, 23(2), 233–250.

Kinoshita, N. & Sheppard, C. (2011). Validating acoustic measures of speech rhythm for second language acquisition. In *Proceedings of the 17th International Congress of Phonetic Sciences*, Hong Kong, 10861089. www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2011/OnlineProceedings/RegularSession/Kinoshita/Kinoshita.pdf.

Ladd, D. R. (2008). *Intonational Phonology*. Cambridge: Cambridge University Press.

Lai, C., Evanini, K., & Zechner, K. (2013). Applying rhythm metrics to non-native spontaneous speech. In P. Badin, T. Hueber, G. Bailly, D. Demolin, & F. Raby, eds., *Proceedings of Speech and Language Technology in Education (SLaTE)*, Grenoble, France, pp. 159–163. www.pure.ed.ac.uk/ws/portalfiles/portal/14978203/laiEtAl2012rhythm.pdf.

Add Reference: Leal Arenas, S. & Ortega-Liebrán, M. (2024). The Prosody of Theticity and Focus in Beginner L2 Spanish. In *Proceedings of Speech F*

Levis, J. (2020). Revisiting the intelligibility and nativeness principles. *Journal of Second Language Pronunciation*, 6(3), 310–328.

Li, A. & Post, B. (2014). L2 acquisition of prosodic properties of speech rhythm: Evidence from L1 Mandarin and German learners of English. *Studies in Second Language Acquisition*, 36(2), 223–255.

Lin, C. Y., Wang, M., Idsardi, W. J., & Xu, Y. I. (2014). Stress processing in Mandarin and Korean second language learners of English. *Bilingualism: Language and Cognition*, 17(2), 316–346.

- Lippi-Green, R. (2012). *English with an Accent: Language, Ideology, and Discrimination in the United States*. New York: Routledge.
- Llisterri, J. & Schwab, S. (2019). Perception of lexical stress in Spanish L2 by French speakers. In M. Gibson & J. Gil, eds., *Romance Phonetics and Phonology*. Oxford: Oxford University Press, pp. 177–190.
- Mennen, I. (2015). Beyond segments: Towards a L2 intonation learning theory. In E. Delais-Roussaire, M. Avanzi, & S. Herment, eds., *Prosody and Language in Contact*. Berlin: Springer, pp. 171–188.
- Mennen, I., Reubold, U., Endes, K., & Mayr, R. (2022). Plasticity of native intonation in the L1 of English migrants to Austria. *Languages*, 7(3), 241.
- Murty, L., Otake, T., & Cutler, A. (2007). Perceptual tests of rhythmic similarity: I. Mora rhythm. *Language and Speech*, 50(1), 77–99.
- Nagao, J. & Ortega-Llebaria, M. (2021). The interaction of micro- and macro-rhythm measures in English and Japanese as first and second languages. In *Proceedings of the 1st International Conference on Tone and Intonation (TAI)*, pp. 273–277. <https://doi.org/10.21437/TAI.2021-56>.
- Nazzi, T., Bertoncini, J., & Mehler, J. (1998). Language discrimination by newborns: Toward an understanding of the role of rhythm. *Journal of Experimental Psychology: Human Perception and Performance*, 24(3), 756–766.
- Nazzi, T. & Ramus, F. (2003). Perception and acquisition of linguistic rhythm by infants. *Speech Communication*, 41(1), 233–243.
- Niebuhr, O., Thumm, J., & Michalsky, J. (2018). Shapes and timing in charismatic speech: Evidence from sounds and melodies. In K. Klessa, J. Bachan, A. Wagner, M. Karpiński, and D. Śledziński, eds., *Proceedings of the 9th International Conference on Speech Prosody*. International Speech Communication Association, pp. 582–586. <https://doi.org/10.21437/SpeechProsody.2018-118>.
- Ohala, J. J. (1984). An ethological perspective on common cross-language utilization of F_0 of voice. *Phonetica*, 41(1), 1–16.
- Ordin, M. & Mennen, I. (2015). Comparison of fundamental frequency in Welsh and English in bilingual speech. In M. Wolters, J. Livingstone, B. Beattie, R. Smith, M. MacMahon, J. Stuart-Smith, & J. M. Scobbie, eds., *Proceedings of the 18th International Congress of Phonetic Sciences (ICPhS 2015)*. Glasgow: Glasgow University. www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2015/Papers/ICPHS0148.pdf.
- Ordin, M. & Mennen, I. (2017). Cross-linguistic differences in bilinguals' fundamental frequency ranges. *Journal of Speech, Language, and Hearing Research*, 60(6), 1493–1506.
- Ordin, M. & Polyanskaya, L. (2015). Acquisition of speech rhythm in a second language by learners with rhythmically different native languages. *Journal of the Acoustical Society of America*, 138(2), 533–544.
- Orrico, R. & D'Imperio, M. (2020). Individual empathy levels affect gradual intonation-meaning mapping: The case of biased questions in Salerno Italian. *Laboratory Phonology*, 11(1), 12.
- Ortega-Llebaria, M. & Colantoni, L. (2014). L2 English intonation: Relations between form-meaning associations, access to meaning, and L1 transfer. *Studies in Second Language Acquisition*, 36(2), 331–353.

- Ortega-Llebaria, M., Gu, H., & Fan, J. (2013). English speakers' perception of Spanish lexical stress: Context-driven L2 stress perception. *Journal of Phonetics*, 41(3-4), 186-197.
- Ortega-Llebaria, M., Nemogá, M., & Presson, N. (2017). Long-term experience with a tonal language shapes the perception of intonation in English words: How Chinese-English bilinguals perceive "Rose?" vs. "Rose." *Bilingualism: Language and Cognition*, 23(2), 207-223.
- Add this reference: Ortega-Llebaria, M., Silva Jr, L., & Nagao, J. (2023). Macro and micro-rhythm in L2 Spanish. *Journal of Phonetics*, 50, 101010.
- Ortega-Llebaria, M. & Wu, Z. (2021). Chinese-English speakers' perception of pitch in their non-tonal language: Reinterpreting English as a tonal-like language. *Language and Speech*, 64(2), 467-487.
- Ortín, R. (2022). Spanish heritage speakers' processing of lexical stress. *International Review of Applied Linguistics in Language Teaching*. <https://doi.org/10.1515/iral-2021-0187>.
- Ortín, R. & Simonet, M. (2022). Phonological processing of stress by native English speakers learning Spanish as a second language. *Studies in Second Language Acquisition*, 44(2), 460-482.
- Ortín, R. & Simonet, M. (2023). Perceptual sensitivity to stress in native English speakers learning Spanish as a second language. *Laboratory Phonology*, 14(1), 1-41.
- Ozaki, Y., Yazawa, K., & Kondo, M. (2017). L2 English speech rhythm of Japanese speakers: An alternative implementation of the Varco metrics. In *Proceedings of the Phonetics Teaching and Learning Conference UCL*, 84-88.
- Parera, A. F. & Lynch, A. (2021). The effects of explicit instruction on written accent mark usage in basic and intermediate Spanish heritage language courses. *Journal of Spanish Language Teaching*, 8(1), 16-31.
- Pellegrino, E., Schwab, S., & Dellwo, V. (2021). Native listeners rely on rhythmic cues when deciding on the nativeness of speech. *Journal of the Acoustical Society of America*, 150(4), 2836-2853.
- Peng, S. H., Chan, M. K., Tseng, C. Y., et al. (2005). Towards a Pan-Mandarin system for prosodic transcription. In S.-A. Jun, ed., *Prosodic Typology: The Phonology of Intonation and Phrasing*. Oxford: Oxford University Press, pp. 230-270.
- Peperkamp, S., Vendelin, I., & Dupoux, E. (2010). Perception of predictable stress: A cross-linguistic investigation. *Journal of Phonetics*, 38(3), 422-430.
- Add this reference: Peskova, A. (2023). L2 Spanish and Italian intonation. BoD-Books on Demand.
- Poeppel, D. & Assaneo, M. F. (2020). Speech rhythms and their neural foundations. *Nature Reviews Neuroscience*, 21(6), 322-334.
- Polyanskaya, L., Busà, M. G., & Ordin, M. (2020). Capturing cross-linguistic differences in macro-rhythm: The case of Italian and English. *Language and Speech*, 63(2), 242-263.
- Prieto, P., del Mar Vanrell, M., Astruc, L., Payne, E., & Post, B. (2012). Phonotactic and phrasal properties of speech rhythm. Evidence from Catalan, English, and Spanish. *Speech Communication*, 54(6), 681-702.
- Prieto, P. & Roseano, P., eds. (2010). *Transcription of Intonation of the Spanish Language*. Munich: Lincom Europa.
- Quené, H., Orr, R., Campbell, N., Gibbon, D., & Hirst, D. (2014). Long-term convergence of speech rhythm in L1 and L2 English. *Social and Linguistic Speech Prosody*, 7, 342-345.
- Qin, Z., Chien, Y. F., & Tremblay, A. (2017). Processing of word-level stress by Mandarin-speaking second language learners of English. *Applied Psycholinguistics*, 38(3), 541-570.

- Ramus, F., Nespor, M., & Mehler, J. (1999). Correlates of linguistic rhythm in the speech signal. *Cognition*, 73(3), 265–292.
- Rao, R. (2019). The phonological system of adult heritage speakers of Spanish in the United States. In S. Colina & F. Martínez-Gil, eds., *The Routledge Handbook of Spanish Phonology*. New York: Routledge, pp. 439–451.
- Rao, R., Ye, T., & Butera, B. (2022). The prosodic expression of sarcasm vs. sincerity by heritage speakers of Spanish. *Languages*, 7(1), 17.
- Rathcke, T., Lin, C. Y., Falk, S., & Bella, S. D. (2021). Tapping into linguistic rhythm. *Laboratory Phonology*, 12(1), article 11, 1–32.
- Romera, M. & Elordieta, G. (2013). Prosodic accommodation in language contact: Spanish intonation in Majorca. *International Journal of the Sociology of Language*, 221, 127–151.
- Romera, M. & Elordieta, G. (2020). Information-seeking question intonation in Basque Spanish and its correlation with degree of contact and language attitudes. *Languages*, 5(4), 70.
- Rosa, J. & Flores, N. (2017). Unsettling race and language: Toward a raciolinguistic perspective. *Language in Society*, 46(5), 621–647.
- Saalfeld, A. K. (2012). Teaching L2 Spanish stress. *Foreign Language Annals*, 45(2), 283–303.
- Sebina, B., Setter, J., & Daller, M. (2021). The Setswana speech rhythm of 6–7 year-old Setswana-English bilingual children. *International Journal of Bilingualism*, 25(3), 592–605.
- Shook, A. & Marian, V. (2016). The influence of native-language tones on lexical access in the second language. *Journal of the Acoustical Society of America*, 139(6), 3102–3109.
- Simonet, M. (2008). Language contact in Majorca: An experimental sociophonetic approach. [Doctoral dissertation, University of Illinois at Urbana-Champaign].
- Soler, J. & Morales-Galvez, S. (2022). Linguistic justice and global English: Theoretical and empirical approaches. *International Journal of the Sociology of Language*, 2022 (277), 1–16.
- Taguchi, N., Hirsch, K., & Kang, O. (2022). Development of prosody in pragmatics meaning-making: Prosodic changes in L2 speech acts and contributing factors. *Studies in Second Language Acquisition*, 44(3), 843–858.
- Van Maastricht, L., Krahmer, E., Swerts, M., & Prieto, P. (2019). Learning direction matters: A study on L2 rhythm acquisition by Dutch learners of Spanish and Spanish learners of Dutch. *Studies in Second Language Acquisition*, 41(1), 87–121.
- Wang, X. (2021). Beyond segments: Towards a lexical model for tonal bilinguals. *Journal of Second Language Studies*, 4(2), 245–267.
- Wang, X., Wang, J., & Malins, J. (2017). Do you hear “feather” when listening to “rain”? Lexical tone activation during unconscious translation: Evidence from Mandarin-English bilinguals. *Cognition*, 169, 15–24.
- Add: Ward, N. G. (2019). *Prosodic patterns in English conversation*. Cambridge University Press.
- White, L. & Mattys, S. L. (2007). Calibrating rhythm: First language and second language studies. *Journal of Phonetics*, 35(4), 501–522.
- Wiener, S. & Goss, S. (2018). Perceptual assimilation of non-native prosodic cues: Cross-linguistic effects of lexical F0 learning. In K. Klessa, J. Bachan, A. Wagner, M. Karpiński, and D. Śledziński, eds., *Proceedings of the 9th International Conference on Speech Prosody*. International Speech Communication Association, pp. 947–951. <https://doi.org/10.21437/SpeechProsody.2018-191>.

- Wiener, S. & Goss, S. (2019). Second and third language learners' sensitivity to Japanese pitch accent is additive: An information-based model. *Studies in Second Language Acquisition*, 41(4), 897–910.
- Winter, B., Oh, G. E., Hübscher, I., et al. (2021). Rethinking the frequency code: A meta-analytic review of the role of acoustic body size in communicative phenomena. *Philosophical Transactions of the Royal Society B*, 376(1840), 20200400.
- Xu, Y., Chen, S.-W., & Wang, B. (2012). Prosodic focus with and without post-focus compression (PFC): A typological divide within the same language family? *Linguistic Review*, 29, 131–147.
- Zubizarreta, M. L. (1998). *Prosody, Focus, and Word Order*. Linguistic Inquiry Monographs, vol. 33. Cambridge, MA: MIT Press.
- Zubizarreta, M. L. & Nava, E. (2011). Encoding discourse-based meaning: Prosody vs. syntax. Implications for second language acquisition. *Lingua*, 121(4), 652–669.

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